

# Functional Requirements Document (FRD)

## Project: AI Study Planner

AI Study Planner is a SaaS-style web application built using React and Firebase. The platform helps students organize their studies intelligently by generating AI-based study schedules, tracking progress, managing notes, and improving productivity through features like Pomodoro timers and streak systems.

### 1. Project Objectives

- Help students manage study time efficiently.
- Generate personalized AI-powered study schedules.
- Track learning consistency using streak systems.
- Improve productivity using Pomodoro sessions.
- Provide analytics and insights for study progress.

### 2. Target Users

- College Students
- School Students
- Competitive Exam Aspirants
- Self-Learning Developers
- Working Professionals preparing for certifications

### 3. Core Features

- User Authentication
- Subject & Topic Management
- AI-generated Study Schedule
- Calendar Planner
- Daily Study Streaks
- Pomodoro Timer
- Notes Management
- Progress Analytics Dashboard
- Dark / Light Theme
- Responsive Design

### 4. Functional Requirements

| Module         | Description                                                                           |
|----------------|---------------------------------------------------------------------------------------|
| Authentication | Users can register, login, logout, and reset passwords using Firebase Authentication. |

|                       |                                                                                 |
|-----------------------|---------------------------------------------------------------------------------|
| Subject Management    | Users can add, edit, delete, and organize study subjects and topics.            |
| AI Schedule Generator | Generate smart study plans based on available hours, deadlines, and priorities. |
| Calendar Planner      | Display study plans and upcoming tasks in calendar view.                        |
| Pomodoro Timer        | Track focused study sessions with configurable timers.                          |
| Notes Section         | Users can create and manage study notes.                                        |
| Analytics Dashboard   | Display charts for study hours, completion percentage, and consistency.         |
| Theme Settings        | Switch between dark mode and light mode.                                        |

## 5. Non-Functional Requirements

- Responsive design for desktop, tablet, and mobile.
- Fast loading performance using React and Vite.
- Real-time database updates using Firebase Firestore.
- Secure authentication and protected routes.
- Scalable frontend architecture.
- Accessibility-friendly UI components.

## 6. Technology Stack

| Technology              | Purpose             |
|-------------------------|---------------------|
| React JS                | Frontend Framework  |
| Firebase Authentication | User Authentication |
| Firebase Firestore      | Realtime Database   |
| Firebase Storage        | File Storage        |
| Tailwind CSS            | UI Styling          |
| Redux Toolkit / Zustand | State Management    |
| Recharts                | Analytics Charts    |
| Framer Motion           | Animations          |

## 7. User Flow

1. User signs up/login.
2. User adds subjects and study goals.
3. AI generates personalized study schedule.
4. User tracks tasks in calendar planner.
5. User studies using Pomodoro timer.
6. Progress and streaks update automatically.
7. User views analytics dashboard and improves consistency.

## 8. Future Enhancements

- AI chatbot study assistant
- Collaborative group study rooms
- Push notifications

- Offline support (PWA)
- Voice notes
- AI exam prediction analytics

## 9. Conclusion

The AI Study Planner project is a modern React + Firebase SaaS application that combines productivity tools, analytics, and AI-based planning into a single platform. The project demonstrates strong frontend architecture, dashboard design, state management, realtime features, and user-centric UX design.